

ASHISH PATIL

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SUMMARY

Software Engineer with experience building distributed, event-driven systems, backend infrastructure, and full-stack applications using Python and modern software engineering practices. Built and shipped two published PyPI packages with 14,000+ combined downloads. Skilled in backend engineering, distributed systems, REST APIs, cloud technologies, and full-stack development, with hands-on experience in agentic AI systems and designing scalable, production-ready software.

EDUCATION

Pimpri Chinchwad College of Engineering, Pune

Aug 2023 – June 2027

B.Tech in Computer Engineering

CGPA: 7.84 / 10

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database & Management Systems, Machine Learning, Deep Learning, Generative AI, Cloud Computing, Distributed Systems

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL

AI / ML: LangChain, LangGraph, LangSmith, RAG, LLM, Agentic Systems, OpenAI, Anthropic, Gemini, Groq

Backend: Node.js, Express.js, FastAPI, REST API

Frontend: React.js, Next.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Firebase, Supabase

Cloud & DevOps: AWS(EC2, S3, RDS, DynamoDB, API Gateway), Google Cloud, Docker, Git, Linux, CI/CD

AI Tools: Claude Code, Antigravity, GitHub Copilot, Cursor, Codex

PROJECTS

Guardian-Runtime: AI Agent Security Proxy

Python · FastAPI · Pydantic · PyPI

10,000+ PyPI downloads · 20 GitHub Stars · Published on PyPI

- Built a local, distributed security proxy and firewall for autonomous AI coding agents, intercepting every LLM prompt and response at the transport layer to enforce FinOps budgets, prevent credential leaks, and compress token payloads entirely on-device before any data leaves the machine.
- Architected a two-layer token optimization engine achieving **40–70% reduction in output token costs**; built pluggable FastAPI/Pydantic middleware supporting OpenAI, Gemini, and Claude with configurable per-session budget limits and zero-code provider switching.

Scankii: AI Agent Security Scanner

Python · PyPI · GitHub Actions

4,000+ PyPI downloads · Published on PyPI

- Built the first open-source security scanner for AI agent skills that jointly analyzes source code and natural-language instructions to detect cross-modal credential leaks, prompt injection, MCP supply-chain attacks, and other AI-specific vulnerabilities before deployment.
- Achieved **100% precision, 68.2% recall, F1 81.1%** on SkillLeakBench (Chen et al., ASE 2026) without dynamic sandbox execution; ships `scankii.runtime` for auto-redacting credentials before they reach the LLM context window, a GitHub Action with SARIF upload, and a pre-commit hook.

CET Buddy: College Predictor Platform

JavaScript · Python · Data Engineering

- Engineered a fully client-side college prediction engine parsing **2,20,000+ MHT-CET cutoff records (2023–2026)** fully in-browser with sub-second filter response across percentile, category, branch, and college dimensions, with paginated results and one-click PDF export.
- Built a Python data pipeline normalizing government-published datasets across 3 academic years; scaled to **1,000+ users** on a mobile-first UI with **~80% reduction in manual research time** for Maharashtra students, with zero server costs and zero paid acquisition.

KodeBattle: Real-Time Competitive DSA Platform

React · Node.js · WebSocket · Firebase

- Architected a real-time, distributed 1v1 competitive programming platform with **event-driven game state synchronization** over persistent WebSocket connections; deterministic result delivery across various matchmaking modes.
- Implemented an Elo-style KOD rating system with persistent match history, real-time analytics (rating graphs, activity heatmaps via Firestore), and an anti-cheat layer with fullscreen enforcement and 3-strike automated forfeit logic.

OPEN-SOURCE CONTRIBUTIONS & PUBLICATIONS

- Merged Contributor: claude-skills (23.4k+ GitHub stars)** - Authored and merged two engineering skills, `minimalist` and `strict-api`, improving AI coding agent reliability and preventing hallucinated API usage.
- Vibe Coding Essentials**, a workflow framework for AI-assisted development. 4 operating modes, 25 golden rules and merged contributor to **Awesome Vibe Coding Resources (235+ GitHub stars)**
- Research: "Academic Certificate Verification using AI and Blockchain: A Survey and Conceptual Framework"** - Proposed a multi-layered architecture integrating OCR, CNN-based forgery detection, and permissioned blockchain (Hyperledger Fabric) for real-time academic credential verification; conducted comparative analysis of 12+ existing verification techniques.